Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

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Abstract

Learned water isotherms are used to predict uptake in porous frameworks at unmeasured temperatures, yet nothing in a flexible model prevents uptake that falls as pressure rises. In a pre-registered test, we asked whether building thermodynamics into the model helps. The model learns an adsorbed-phase potential whose derivative is the isotherm, so that for every parameter value uptake rises with pressure, the Henry slope is finite and the isosteric heat is defined. On 13 metal-organic and covalent organic frameworks, predicting the highest or lowest measured temperature, none of its 26 extrapolations was unphysical, against 9 for the same model trained with penalties. Its accuracy could not be separated from the penalty model's (error ratio 1.10, 95 % interval 0.93 to 1.65; the pre-registered cost below 25 % was not shown) or from characteristic-curve, neural-network and dual-site Sips models. With two training temperatures, equally good fits disagreed by two orders of magnitude at a third. On synthetic truths with exact heats, its isosteric heat was always defined and 18 to 146 times more accurate than an unconstrained network's. Built-in constraints secure admissibility; accuracy is limited by the temperature structure and the number of measured temperatures.

Introduction

Water sorbents built from metal-organic frameworks (MOFs) and covalent organic frameworks (COFs) are studied for atmospheric water harvesting[1, 2] and for adsorption heat pumps and chillers[3, 4]. These devices run between temperatures: a MOF-801 cooling cycle adsorbs at 30 °C and regenerates at 80 to 85 °C[5]. Its design needs the uptake at temperatures where no isotherm was measured, and the isosteric heat, which enters the cycle model[4]. Both rest on few temperatures. A practical guide asks for isotherms spaced by 20 K or less for a Clausius-Clapeyron heat and advises against extrapolation beyond the measured range[6]. For Mg-SU-102, isotherms were measured from 15 to 55 °C, but only those from 25 to 40 °C had enough points for the heat[7].

The standard temperature model for water in MOFs is the characteristic curve: the uptake is written as a function of the adsorption potential $A = RT \ln(p_{\mathrm{sat}}/p)$, the isotherms of all temperatures are expected to fall on one curve, and that curve is used to model cycles[8]. Within this approach the isosteric heat is the vaporisation enthalpy plus A[4]. Invariance is usually shown by overlaying curves, without an error number, as for MIL-160 from 20 to 50 °C[1] and Al-fumarate from 298 to 333 K[9]. The Mg-SU-102 curves shift slightly towards lower energies as the temperature rises[7], and MOF-303 loses a large part of its capacity between 25 and 45 °C[10]. Statistical thermodynamics gives a characteristic curve when added sorbate contributes purely energetically, and a temperature-dependent isotherm when sorbate number correlates with entropy[11]. The other common route, Langmuir or Sips sites with temperature-dependent affinities fitted to all temperatures at once[4, 12], has its own defects. Sips and Freundlich terms have no finite slope at zero pressure and their heats diverge at low coverage,

dual-site Langmuir heats can show an unphysical maximum, and models that fit equally well disagree on the heat wherever they extrapolate[12].

Machine-learned isotherm models add flexibility but little thermodynamics. In two recent examples temperature is a plain input: one network predicted methane isotherms of 17,644 MOFs from 278 to 338 K and was extrapolated to 268 and 358 K[13]; another predicts water isotherms from pressure and structural properties[14]. Where heats appear, they are separate outputs. A neural ordinary differential equation learns $\mathrm{d}q/\mathrm{d}\ln p$ at one temperature, observes monotonicity without imposing it, and predicts the heat with a second network[15]. A zeolite study joined a learned loading at 300 K and an independently learned heat through an integrated Clausius-Clapeyron step with a heat that depends on loading but not on temperature[16]. When physics enters, it enters as a penalty. A network for hydrogen sorption adds monotonicity as a weighted loss term and reports 98.6 % monotonicity satisfaction, so 1.4 % of held-out cases violate it[17]. A penalty-trained network is not provably monotone; monotonicity can only be certified afterwards[18].

A different construction is possible. The grand (immersion) potential of the adsorbed phase, $\Omega = -RT \int n \,\mathrm{d}\ln p$, has the isotherm as its derivative with respect to $\ln p$[19, 20]. A model can learn this one scalar as a function of $\ln p$ and T. If it is convex in $\ln p$, the isotherm is monotone for every parameter value, and input-convex networks give convexity by construction[21]. A potential linear in p at low pressure gives a finite Henry slope, which Sips terms lack[12]. The isosteric heat then follows by Clausius-Clapeyron from second derivatives of the same function and is defined wherever $\mathrm{d}q/\mathrm{d}\ln p$ is positive. Neighbouring fields already learn potentials this way. A learned free energy gives fluid properties as its derivatives, with exact Maxwell relations[22]. In mixture thermodynamics, a graph network for the excess Gibbs energy replaced a Gibbs-Duhem penalty[23] and needs no extra loss terms[24], and hard-constraint networks build the Gibbs-Duhem relation into the architecture[25] and take the excess enthalpy from the temperature derivative of a learned excess Gibbs energy[26]. In adsorption, a learned density functional has produced an isotherm of a Lennard-Jones fluid in a model pore[27]. In the sources we searched, no model learns such a potential from measured isotherms of real sorbents or tests it on water.

Hard constraints make a model admissible, but not necessarily accurate. For phase-transition enthalpies, the error of the predicted sublimation enthalpy rose by more than 10 $\mathrm{kJ\,mol^{-1}}$ under a hard thermodynamic constraint, while a soft constraint kept the accuracy[28]. Two parts of the construction can also act separately: the shape constraint fixes monotonicity and the Henry limit, while the temperature structure (a van't Hoff affinity per term, a free dependence on $1/T$, or a characteristic curve) decides how the isotherm moves between temperatures. The open question is which of the two, if either, makes extrapolation in temperature better than penalties, generic networks and classical models.

The present study therefore asks whether hard constraints on a learned adsorbed-phase potential make its temperature extrapolation of measured water isotherms better than penalties, generic networks and classical models. It does so in a pre-registered factorial design that crosses the shape constraint (none, penalty or hard) with the temperature structure (free in $1/T$, van't Hoff per term, or a temperature-invariant characteristic curve). The learned potential is the cell with hard constraints and van't Hoff terms. It is compared with its penalty and unconstrained twins, the hard-constrained cells of the other two temperature structures, a generic neural network and a dual-site Sips model with van't Hoff affinities. The data are 13 real sets of water isotherms measured at three or more temperatures, 5 on MOFs and 8 on COFs. Each model predicts a material's highest and lowest temperatures from the others. The design, tests and words of every verdict were fixed before any confirmatory fit, including words for a classical model that wins and for models equivalent within 25 %. Every defect found in the design, code or data is recorded in a public correction ledger. None of the 13 materials has a calorimetric water heat; our search found one verified value for a MOF with multi-temperature isotherms, a single integral heat for MOF-801[29], and none for any COF. We therefore test the heat on synthetic

truths with exact heats, built on each material's own temperatures and pressures. The study makes four contributions. (i) A learned adsorbed-phase potential whose isotherm is monotone in pressure, has a finite Henry slope and has a defined isosteric heat for every parameter value, with a van't Hoff temperature law per term (Methods). (ii) A test of that model against penalty, unconstrained, generic and classical models whose design, code and verdict words were frozen before any confirmatory fit and scored by two independent analysis codes. (iii) The confirmatory result that the hard constraints removed every violation (0 of 26 extrapolations, against 9 for the penalty model) while their accuracy cost could neither be excluded beyond the margin nor detected (error ratio 1.10, 0.93 to 1.65). (iv) Descriptive and synthetic evidence that the temperature structure and the number of measured temperatures, more than the shape constraints, limit temperature extrapolation and the isosteric heat.

Results

The design and the run

The models form a factorial of shape constraint (none, penalty, hard) and temperature structure (free in $1/T$, a van't Hoff affinity per term, a characteristic curve), and P is its hard van't Hoff cell (Fig. 1). Each material is fitted in three tasks: E-high and E-low hold out the highest or the lowest temperature, and I-mid holds out a middle one. The confirmatory run produced 432 fits: 11 cells on 13 materials and three tasks, and the Kirchhoff variant P-K on the one material with at least four training temperatures, as pre-registered. Every fit was started from 5 seeds, and zero fits failed. The pre-registered analysis and an independent scorer written from the frozen text alone gave the same ratios, intervals, adjusted p values and admissibility classes.

Accuracy of temperature extrapolation (H1)

None of the four confirmatory comparisons separated P from its comparator (Fig. 2a). Against the penalty model V-pen, the same function class with the shape conditions as a training penalty, the ratio of extrapolation errors was $R = 1.10$ (exact 95 % interval 0.93 to 1.65; Holm-adjusted $p = 0.76$). The pre-registered claim, that the hard constraints cost less than 25 % in accuracy, required the upper bound to fall below 1.25; it did not, so the claim is not supported (the upper 98.75 % bound, which applies the family correction, is 3.08). Under equal accuracy the criterion would have been met with probability 0.51 to 0.74 at the spread of per-material differences assumed in the power analysis, 0.158 decades (Supplementary Note 1); the observed spread was 0.30 decades, about twice that. The interval includes both a small advantage for P and a substantial one for V-pen. Against the hard characteristic curve A-hard, $R = 1.51$ (0.84 to 2.87); against the unconstrained network F-none, $R = 0.63$ (0.40 to 1.22); against dual-site Sips with van't Hoff affinities (C1), $R = 1.23$ (0.92 to 1.96). All four Holm-adjusted p values were 0.76, and each verdict reads "inconclusive at this sample size".

One material dominates the spread. On COF-SO_3H, a COF carrying sulfonic acid groups, P's extrapolation error was 10.9 times that of V-pen, 9.9 times that of A-hard, 6.2 times that of F-none and 8.6 times that of C1, the largest ratio in every comparison (Supplementary Table S2). On the other materials the ratios scatter about one: P was more accurate than V-pen on 6 of the 13 materials, than F-none on 8, than C1 on 6 and than A-hard on 5. Split by class (estimates only, 5 MOFs and 8 COFs), the ratio against V-pen was similar for the two ($R = 1.09$ and 1.10), while the characteristic curve did much better than P on the COFs than on the MOFs ($R = 2.67$ against 0.69); the generic network and C1 were also relatively better on the COFs (Supplementary Table S7).

The descriptive comparison with the free temperature network F-hard, which shares P's shape constraints but has no temperature law, favoured P ($R = 0.49$, 0.33 to 0.85, unadjusted $p = 0.027$; smaller error on 11 of 13 materials). The synthetic worlds showed that this comparison

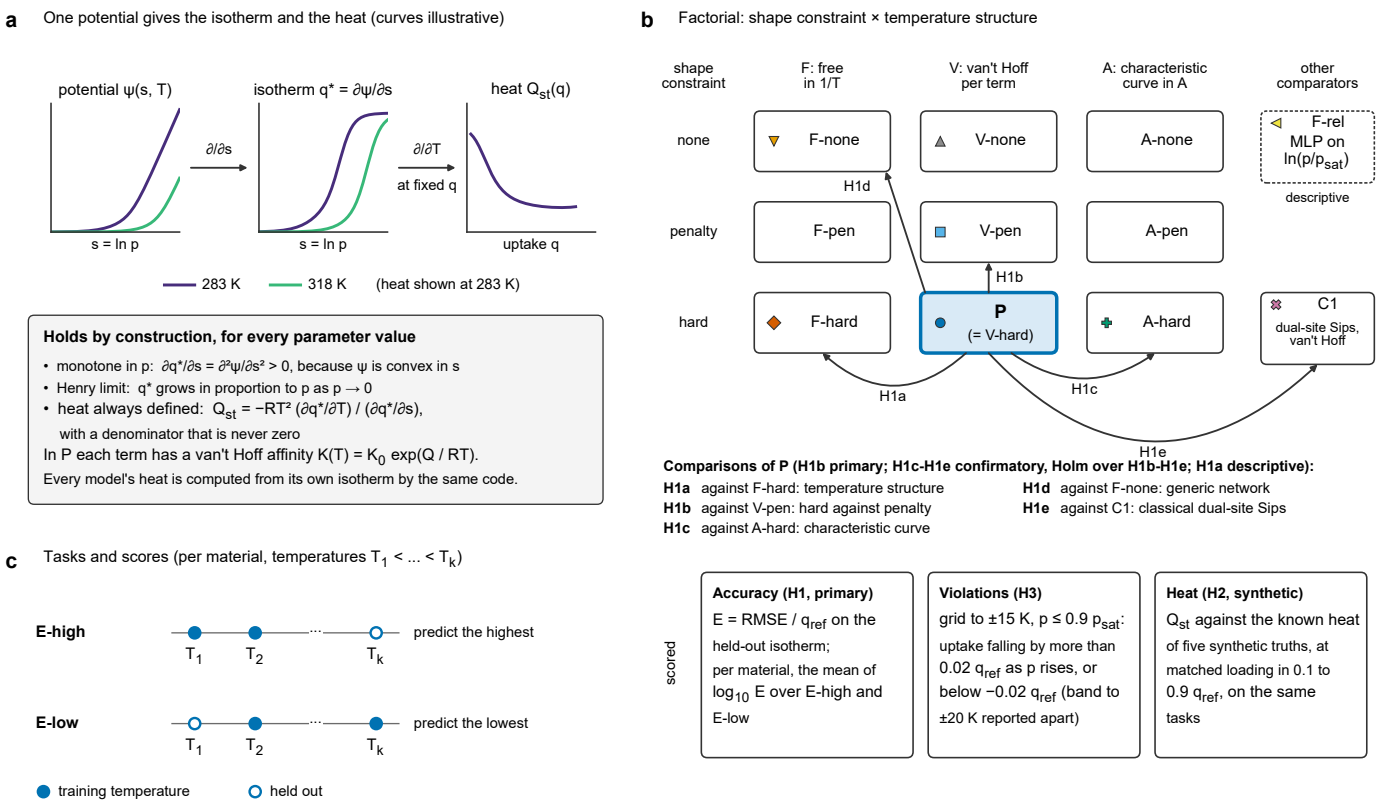


Figure 1: **Model, factorial and tasks. a**, The learned potential $\psi(s, T)$ gives the isotherm as its derivative in $s = \ln p$ and the heat from the isotherm; the three properties in the box hold for every parameter value. **b**, The factorial of shape constraint and temperature structure, the two further comparators and the comparisons of P with their evidence labels. **c**, The extrapolation tasks and the three scores.

cannot fail when only two temperatures are available for training (Supplementary Note 1), so it is reported as a description and not as a test of the temperature law. On the two materials with at least three training temperatures, where the free network has more to learn from, P was also more accurate ($R_m = 0.28$ for Mg-SU-102 and 0.49 for Zn_2(bptc)).

Admissibility of the extrapolated isotherms (H3)

The hard constraints did what they guarantee, and the penalties did not (Fig. 2b). No extrapolation fit of P, F-hard, A-hard or C1 had a violation inside the domain (0 of 26 each; upper 95 % bound 0.13), and the strict counts that must be zero by construction were zero for P, F-hard, A-hard and P-K (0 points), so the bug check passed. The penalty model V-pen violated in 9 of 26 fits (Wilson interval 0.19 to 0.54), no fewer than the unpenalised V-none (8 of 26). The pre-registered prediction, at least one violating fit for V-pen and none for P, was therefore met. Without the van't Hoff structure the penalty helped little: F-pen violated in 13 fits and A-pen in 16, against 26 for F-none and 15 for A-none, and the relative-pressure network F-rel violated in 22. With no tolerance and over the whole grid, V-pen had 11 violating fits (Supplementary Table S3).

Isosteric heat on synthetic truths (H2)

The fifteen heat tests compare P with V-pen, A-hard and F-none on each truth family, sampled on the confirmatory materials' own temperatures and pressures, with no direction predicted

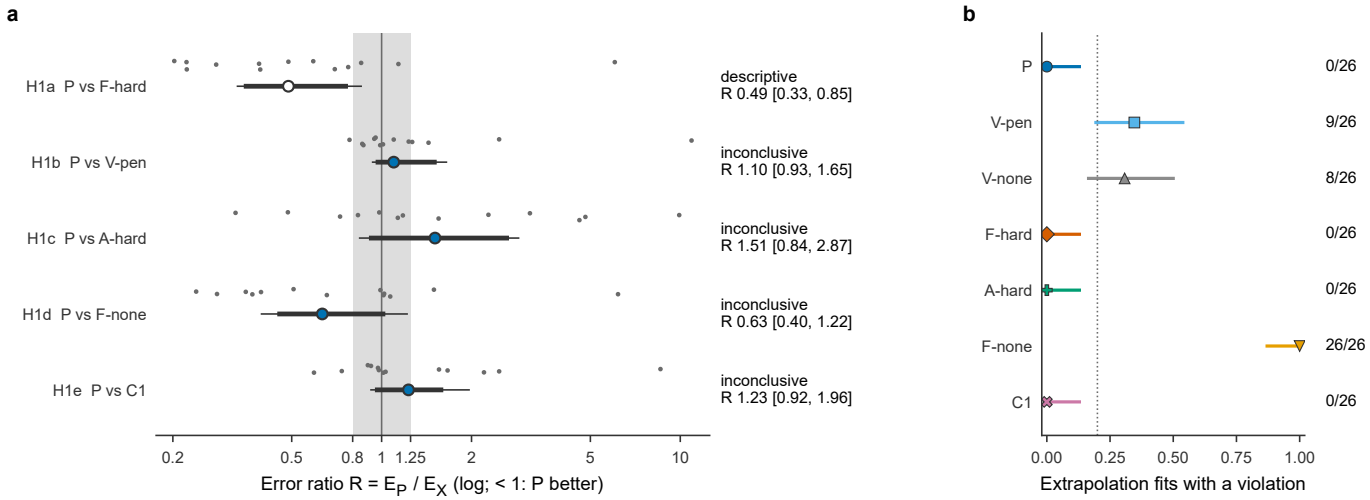


Figure 2: **Confirmatory verdicts.** **a**, Ratio of extrapolation errors $R = E_P/E_X$ (log scale; below one, P is more accurate): Hodges-Lehmann estimate with exact 95 % (thin) and 90 % (thick) intervals over the 13 materials; dots, single materials; shading, the equivalence band 0.8 to 1.25. H1a is descriptive. **b**, Fraction of the 26 extrapolation fits (E-high and E-low) with a violation inside the domain, with Wilson 95 % intervals; dotted line, the pre-registered 20 % threshold.

(Fig. 3 and Supplementary Table S5). P's heat was never undefined: zero of its tasks failed, against 28 of the 130 tasks of the unconstrained network. Over all tasks the median root-mean-square heat error was 0.73 kJ mol^{-1} for P, 0.70 for V-pen, 3.06 for A-hard and 9.9 for F-none.

Against F-none, P's heat error was smaller in every family, by factors of $R = 0.007$ to 0.056. Against A-hard it was smaller in the four families outside the characteristic-curve family ($R = 0.023$ to 0.41); on the Dubinin-Astakhov truth, which lies inside A-hard's family and whose heat is exactly the vaporisation enthalpy plus A, A-hard was far better ($R = 114$). Against the penalty model the differences were small: P was better on the strong-site family ($R = 0.66$, 0.53 to 0.84), better by an uncertain amount on the two-site family with a heat capacity ($R = 0.83$, 0.74 to 0.98), and equivalent on the Dubinin-Astakhov family ($R = 0.98$); on the Fowler-Guggenheim and capacity-loss families the comparison was inconclusive, and V-pen's heat failed on three materials of the Fowler-Guggenheim family.

Descriptive and exploratory analyses

Absolute errors. Over the 26 extrapolation fits, the median error was 6.0 % of q_{ref} for A-hard, 8.4 % for C1, 8.5 % for V-pen, 9.7 % for P and 10.8 % for F-none. Predicting a middle temperature is easier: P's median error in I-mid was 4.6 %, and in that task P was equivalent to V-pen ($R = 1.02$, 0.91 to 1.20) and to C1 ($R = 1.00$, 0.92 to 1.13) within the 25 % margin, and slightly but not clearly more accurate than A-hard ($R = 0.78$, 0.62 to 1.02).

The constraints at equal structure. Against V-none, the same model without its shape constraints, P's extrapolation error ratio was $R = 1.14$ (0.94 to 1.50), and its training loss was higher by a median factor of 2.3 over the 39 fits: the constraints cost some fit to the training temperatures and bought no clear gain in extrapolation. Against the relative-pressure network F-rel, $R = 1.13$ (0.70 to 1.89). On Mg-SU-102, the one material with enough temperatures for the Kirchhoff variant, P's error was 0.74 times that of P-K. The per-material differences did not

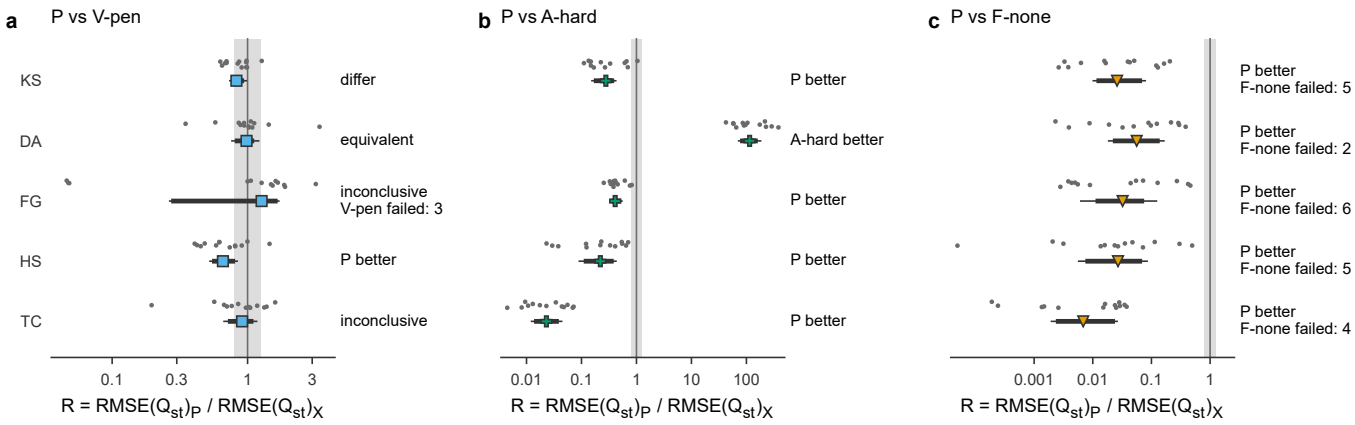


Figure 3: **Isosteric heat on synthetic truths (H2).** Ratio of the root-mean-square error of the heat at matched loading, P over the comparator (log scale; below one, P is closer to the exact heat), for each of the five truth families: Hodges-Lehmann estimate with exact 95 % (thin) and 90 % (thick) intervals over the 13 materials; dots, single materials (a failed task is ranked worst); shading, 0.8 to 1.25. KS, two-site isotherm with a heat capacity; DA, Dubinin-Astakhov; FG, Fowler-Guggenheim; HS, strong hydrophilic sites; TC, capacity falling with temperature (Supplementary Note 2). **a**, against V-pen; **b**, against A-hard; **c**, against F-none.

grow with the distance of the held-out temperature from the training temperatures (Spearman correlations −0.16 to 0.26 across the four comparisons).

Seed spread. The five seeds of a fit usually agreed: the median range of $\log_{10} E$ over seeds was 0.04 decades for P, 0.01 to 0.13 for the other cells with a van't Hoff or characteristic-curve structure, and 0.18 to 0.67 for the free networks (Supplementary Table S4). The exceptions matter. For P on COF-SO_3H in E-high, the five seeds fitted the two training temperatures equally well and gave held-out errors from 0.035 to 7.7; the pre-registered rule, the lowest training loss, selected a seed with $E = 4.04$. Across P's 26 extrapolation fits the rule selected the seed with the best held-out error 6 times and the worst 9 times, no better than chance: with two training temperatures the training loss does not identify the temperature dependence.

Two exploratory checks. The following analyses were not pre-registered; they were added after the verdicts were known, because of the seed spread above (correction ledger, entry X-1). With each fit's error replaced by its median over the five seeds, for every model, the ratio against V-pen was $R = 1.00$ (0.90 to 1.24), inside the non-inferiority margin. With COF-SO_3H left out and the selected seeds kept, it was $R = 1.05$ (0.91 to 1.35), not inside it. The pre-registered verdict stands; the checks show that it turns on how one seed of one fit is chosen.

Sensitivity to the data. The H1 comparisons were repeated on the seven pre-registered dataset variants (Supplementary Table S6). The ratio against V-pen stayed between 0.97 and 1.19; its verdict was inconclusive in five variants and equivalent in two, and the non-inferiority bound fell below 1.25 in one: without the coverage rule, which admits 23 materials including isotherms with occluded markers, the ratio was $R = 1.05$ (0.97 to 1.20). In the same variant P extrapolated better than F-none ($R = 0.59$, Holm-adjusted $p = 0.012$), and the characteristic curve came close to being better than P ($R = 1.52$, 1.06 to 2.19, Holm-adjusted $p = 0.064$). Shifting every digitised isotherm by the largest measured systematic digitisation error changed no verdict except that against F-none in one of the two perturbed copies, where it read "P extrapolates better".

One calorimetric value. For MOF-801, the one material with multi-temperature isotherms and a calorimetric water heat (an integral value of 53.3 $\mathrm{kJ\,mol^{-1}}$ at 30 °C[29]), every model fitted to its five temperatures gave a heat between 48.5 and 54.1 $\mathrm{kJ\,mol^{-1}}$ in the loading window of the measurement (P, 51.0). One integral value cannot rank the models.

Where the step lands. For water in these frameworks the uptake step is the quantity that moves with temperature. Refitting the reported seed of every extrapolation fit of the six models in the H1 comparisons (156 fits, each reproducing its recorded error exactly), the median absolute error in the relative pressure of the step at the held-out temperature was 0.011 for V-pen, 0.013 for C1 and 0.014 for both A-hard and P (0.014), against 0.028 for F-none and 0.059 for F-hard. The structured models placed the step equally well; the free temperature networks did not.

Discussion

What the hard constraints buy. The clearest result is about admissibility, not accuracy. Every extrapolated isotherm of P was monotone and non-negative inside the domain where the penalties were applied, as the construction guarantees, while the penalty model V-pen left violations in 9 of 26 fits, as many as the same model with no penalty at all (8). The penalty weight was tuned, as the pre-registration required, for accuracy on the development materials, and the tuning chose a weight small enough to leave the extrapolations almost as unconstrained as before. This is a general tension rather than a defect of our tuning: a penalty strong enough to remove violations everywhere also pulls the fit away from the data, so a weight chosen for accuracy settles where violations survive. A hard constraint removes the trade. For a process model that integrates an isotherm over a temperature swing, a single non-monotone stretch can make an adsorber appear to release water on loading or produce a negative heat, and such errors are found only if someone looks for them.

What they cost. On accuracy, the test we declared primary did not succeed: the upper bound of the error ratio against V-pen, 1.65, exceeded the 1.25 margin. The data therefore do not exclude that hard constraints cost more than 25 % in extrapolation accuracy. They do not indicate such a cost either. The point estimate is $R = 1.10$; P was more accurate on 6 of the 13 materials; in interpolation the two were equivalent within the margin; and the shortfall rests on one fit whose selected seed was among the worst of its five. The exploratory checks agree with this reading in one direction and not in the other (the median over seeds brings the bound inside the margin, leaving COF-SO_3H out does not), so we report the pre-registered verdict and do not claim non-inferiority. A test of the margin with this power needs either more materials or a smaller spread of per-material differences, which in turn needs more training temperatures per material.

Temperature structure matters more than shape constraints. The contrasts within the factorial point the same way. Free temperature networks had the largest seed spread and the largest errors, and P extrapolated better than the free network with the same shape constraints on 11 of 13 materials, although the synthetic worlds show that this comparison cannot fail with two training temperatures. Among the structured models, the hard characteristic curve had the lowest median extrapolation error (6.0 % of q_{ref}, against 9.7 % for P). The characteristic curve is the standard temperature model for water in MOFs[8], and our comparison, while inconclusive ($R = 1.51$, 0.84 to 2.87), does not suggest that a learned potential with van't Hoff positions beats it for water in these frameworks. A model that combines the potential's guarantees with a characteristic-curve temperature structure is the obvious next candidate; A-hard is close to such a model, and our results support testing it on materials with more temperatures.

Two temperatures do not determine a temperature dependence. Eleven of the 13 materials offer only two training temperatures in the extrapolation tasks. With two temperatures, fits that describe the training data equally well can disagree by two orders of magnitude at a third temperature, as P's seeds did on COF-SO_3H, and the training loss cannot tell them apart. Any learned isotherm model fitted to so few temperatures should report the spread over restarts, not only the best training fit, and claims of temperature extrapolation from two-temperature data deserve the same caution. For experimentalists, the most useful step is cheap: water isotherms at four or more temperatures, with numeric data deposited.

The heat follows the isotherm. A model's isosteric heat is a temperature derivative of its isotherm, so it inherits every weakness of the temperature structure. On synthetic truths with exact heats, the unconstrained network gave heats one to two orders of magnitude worse than P and failed to give a defined heat in 28 of 130 tasks; the characteristic curve was exact on its own family and several times worse elsewhere, because it fixes the heat to the vaporisation enthalpy plus the adsorption potential. P and the penalty model gave heats of similar quality (median errors 0.73 and 0.70 $\mathrm{kJ\,mol^{-1}}$), with P ahead on the families that have a strong-site or heat-capacity contribution. P's median heat error per family was 0.09 to 1.45 $\mathrm{kJ\,mol^{-1}}$, including the two families outside its function class (Fowler-Guggenheim and Dubinin-Astakhov). We make no claim about the heat of any real material, for which calorimetric data are missing.

Limitations. The confirmatory set has 13 materials, the number of MOFs and COFs with water isotherms at three or more temperatures that passed our inclusion rules, and most of the isotherms were digitised from published figures. The digitisation error was measured (Supplementary Note 3) and is small against the extrapolation errors, but occlusion of markers excluded several data sets, and the excluded materials may differ from the included ones. Each material's temperatures span 15 to 40 K, so the extrapolations are short. The models describe one adsorption branch; hysteresis and framework flexibility are outside their scope. No confirmatory material has a calorimetric water heat, so no claim is made that any model's heat is closer to the true heat of a real material. The Henry limit cannot be tested on these data, because no isotherm reaches the Henry regime, so the finite Henry slope of P is a property of the model, not a validated prediction.

Conclusion. Built-in thermodynamic constraints gave admissible extrapolations in every fit, which training penalties did not, at an accuracy cost that this pre-registered test could neither exclude beyond the 25 % margin nor detect. For temperature extrapolation of water isotherms in these frameworks, the results point to the temperature structure of the model and the number of measured temperatures, more than to the shape constraints, as what limits accuracy.

Methods

Pre-registration

The design, the data, the models, the tuning rules, the metrics, the statistical tests and the words of every verdict were fixed in a pre-registration and frozen before any confirmatory fit. The freeze is a public, tagged commit (`prereg-p3a-freeze`) whose manifest holds the SHA-256 of 71 files (the text, the data, the tuned configurations and the fits behind them, the synthetic-world and power results, and every script that builds, fits or scores, including an independently written second scorer) and the package versions; the confirmatory scripts refuse to run if any of them differs. Every defect found in the design, code or data, before or after the freeze, is listed with its cause and fix in a public ledger.

Data

Water adsorption isotherms of MOFs and COFs measured at several temperatures on one sample were collected from the primary literature and from the NIST/ARPA-E adsorption database[30]. The confirmatory data come from published studies[31, 32, 2, 33, 34, 35, 36, 7, 37, 38], and the development data from two further studies and one of the same[39, 40, 36]; Supplementary Table S1 gives the source of each material. A material entered the confirmatory set if, at three or more temperatures, its adsorption isotherm had at least 8 points with $0 < p/p_{\mathrm{sat}} \leq 0.90$, its figure passed the digitisation quality gates, its temperatures were unambiguous, and its curve was covered: no gap above 0.25 in p/p_{sat}, and a first point at $p/p_{\mathrm{sat}} \leq 0.10$ or at an uptake of at most 0.10 q_{ref}. Where a paper gave both measured isotherms and curves derived from them, the measured isotherms were used. The confirmatory set has 13 materials, 5 MOFs and 8 COFs; every decision and its reason is in the inclusion log. Four further materials form a development set used only to choose hyperparameters.

Isotherms published only as figures were digitised before any model was fitted. Vector figures were read from the PDF drawing commands, with marker centres mapped through the axis ticks; raster figures by template matching with the figure's own marker footprint, with tick marks located to sub-pixel precision and an open-marker veto against desorption points. No physical prior was used to filter points. The method was validated against published numeric data. On Mg-SU-102 the vector extraction recovered 198 of 198 points with a root-mean-square uptake error of 7×10^{-6} g g^{-1}. On raster figures of HKUST-1 the root-mean-square uptake error was 0.11 to 0.19 % of the axis span, or 0.0012 to 0.0034 of the isotherm's maximum, with no false detections.

Only adsorption branches with $p/p_{\mathrm{sat}} \leq 0.90$ were used. Relative pressures were converted with the IAPWS saturation pressure, absolute pressures were used as reported, and uptakes were converted to mol kg^{-1}. The scale q_{ref} of a material, its largest uptake in this range over all temperatures, is used only for scoring.

Models

Every model is a function $q(s, T)$ of $s = \ln(p/\mathrm{Pa})$ and temperature, and the isosteric heat of every model is computed by the same code, $Q_{\mathrm{st}} = -RT^2(\partial q/\partial T)/(\partial q/\partial s)$, by automatic differentiation.

The potential model (P) learns the reduced spreading pressure

$$\psi(s,T) = \sum_k \frac{c_k(T)}{g_k}\,\mathrm{softplus}\big(g_k\,(s-\theta_k(T))\big), \qquad q^* = \frac{\partial \psi}{\partial s} = \sum_k c_k(T)\,\sigma\big(g_k\,(s-\theta_k(T))\big), \quad (1)$$

with capacities $c_k > 0$, Henry terms ($g_k = 1$) and cooperative terms ($g_k \geq 1.5$), van't Hoff positions $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}})$ and an adsorbate-expansion factor. For every parameter value, q^* increases strictly with p, q^*/p tends to a positive Henry constant as $p \to 0$, and the heat is defined everywhere.

The models form a factorial of shape constraint (none, penalty, hard) and temperature structure: a free network in $1/T$ (F), van't Hoff per term (V), or the temperature-invariant characteristic curve in A (A). In the V and A columns, the unconstrained and penalised models have the function class of P or of the hard characteristic curve without the constraints, the same initial function and the same tuned structure; the penalty models add monotonicity and (except for the characteristic curve) a scale-free Henry-limit penalty on collocation points that reach the held-out temperature. A dual-site Sips model with van't Hoff affinities (C1) is the classical comparator. A multilayer perceptron on $(\ln p/p_{\mathrm{sat}}, 1/T)$ and a Kirchhoff variant of P with a heat capacity per term are reported descriptively.

Fitting and tuning

Each model was fitted by Adam followed by L-BFGS on the mean squared error of q scaled by the largest training uptake, in double precision, from 5 seeds; the fit with the lowest training loss was kept, never selected on held-out error. Every scale of a fit came from the training temperatures only. Hyperparameters were chosen on the development set alone, in two stages: the models that own a structure tried up to 12 configurations each, and the unconstrained and penalised models inherited the chosen structure, the penalty models tuning only the penalty weight over 10^{-4} to 100.

Tasks and outcomes

For a material with temperatures $T_1 < \cdots < T_k$, the task E-high trains on $T_1, \ldots, T_{k-1}$ and predicts T_k; E-low trains on $T_2, \ldots, T_k$ and predicts T_1. The primary outcome is the held-out error $E = \mathrm{RMSE}(q_{\mathrm{pred}} - q_{\mathrm{obs}})/q_{\mathrm{ref}}$, summarised per material as the mean of $\log_{10} E$ over the two tasks. Physical violations were counted on a dense grid: a fit violated if an isotherm fell by more than 2 % of q_{ref} below its running maximum, or went below -2 % of q_{ref}, inside the domain where the penalties act; violations were counted over the fits of the two extrapolation tasks. Because no material here has a calorimetric water heat, the heat was tested on synthetic isotherms with exact heats, built on every material's own temperatures and pressures from five truth families, with noise of 0.4 % of q_{ref}.

Isosteric heat on synthetic truths

The five truth families are a Langmuir site plus a cooperative Sips site with a heat capacity, a Dubinin-Astakhov curve, a Fowler-Guggenheim isotherm with lateral attraction, strong hydrophilic sites with a heat that falls steeply with loading, and a Sips isotherm whose capacity falls with temperature (Supplementary Note 2). Each family is sampled on every confirmatory material's own temperatures and pressure grid, with shape parameters fitted to that material's middle-temperature isotherm and the thermodynamic parameters fixed in advance. P, V-pen, A-hard and F-none were fitted to the E-high and E-low tasks of every truth. The primary metric compares the heats at equal loading: at each held-out state where the model's own loading lies between 0.1 and 0.9 q_{ref}, the model's Q_{st} is compared with the truth's Q_{st} at the same loading and temperature, found by inverting the monotone truth, and the root-mean-square difference is taken in $\mathrm{kJ\,mol^{-1}}$. A task fails, and is ranked worst, if any compared heat is undefined or not finite, if fewer than 3 states fall in the window, or if the truth cannot reach the model's loading. The fifteen comparisons (three models against P in five families) are Holm-corrected as one family. The single calorimetric value found for a material with multi-temperature isotherms, MOF-801[29], is used only in a descriptive check; no claim about the heat of a real material is made.

Admissibility

For each model the violating fits are counted over the 26 fits of the two extrapolation tasks, with a Wilson 95 % interval; the counts over all tasks, adding the interpolation task, are reported beside them. The words follow a table fixed in advance: at least 20 % of fits when the lower bound reaches 20 %, no violations detected when no fit violates, fewer than 20 % when the upper bound is below it, and an uncertain rate otherwise. The pre-registered prediction is that penalties leave violations that the hard constraints remove: V-pen has at least one violating fit and P has none. It concerns V-pen at the penalty weight chosen for accuracy on the development set ($\lambda = 0.001$), not penalties at every weight. By construction P cannot violate; a violating fit of P would be a bug and halts the analysis.

Descriptive and sensitivity analyses

Fixed in advance and reported without a hypothesis: the interpolation task; P against V-none (the pure effect of the constraints at equal structure), against F-hard (rigid against free temperature dependence), against the multilayer perceptron and against the Kirchhoff variant of P; MOFs and COFs separately; the error against the distance of the held-out temperature from the nearest training temperature; and the spread of the characteristic curve across temperatures. The H1 comparisons were repeated on the datasets given by earlier versions of the coverage rule and by a 7-point rule (6, 9, 23 and 14 materials), on Co_2Cl_2(BTDD) from its characteristic curves, and on two perturbed copies of the data in which every digitised isotherm is shifted in p/p_{sat} by ± 0.002, the largest systematic shift measured in the digitisation benchmark.

Analyses added after the verdicts

Three analyses were specified only after the confirmatory verdicts were known and are listed in the correction ledger (entries X-1 and X-2). The pre-registration requires the seed spread of the held-out error to be reported without fixing its form; we report, per model, the median over its extrapolation fits of the range of $\log_{10} E$ over the 5 seeds, and how often the seed with the lowest training loss had the best or the worst held-out error. Two exploratory checks of the primary comparison followed from it: every fit's error replaced by its median over the seeds, for every model, which uses no held-out information to choose among seeds; and the material with the largest paired difference left out. Both reuse the frozen analysis code. The step-position error, declared in advance but without a frozen script, needs the fitted curves, which the run records do not keep: the selected seed of each extrapolation fit of the models in the confirmatory comparisons was refitted with the frozen code and configuration, and the refit had to reproduce the recorded error. The step position of a curve at the held-out temperature is the p/p_{sat} at which it first reaches half of its own uptake at $p/p_{\mathrm{sat}} = 0.90$.

Software

All fits ran in double precision on a CPU with Python 3.12.10, PyTorch 2.12.0, NumPy 2.4.6, SciPy 1.17.1 and pandas 3.0.3; the versions are part of the frozen manifest.

Statistics

The unit is the material ($n = 13$). For each comparison of P with a model X, $d_m = e_m(\mathrm{P}) - e_m(\mathrm{X})$ was tested with the exact two-sided Wilcoxon signed-rank test, with Holm correction over the four confirmatory comparisons (penalty model, characteristic curve, generic network, dual-site Sips). The effect is the Hodges-Lehmann estimate with exact distribution-free 95 % and 90 % intervals, reported as the error ratio $R = 10^d$. A failed fit was ranked worst, never dropped. The words of each verdict follow a table fixed in advance, with a smallest effect of interest of 25 % ($R = 0.8$ or 1.25). The one declared primary claim, that the hard constraints cost no more than 25 % in accuracy against the penalty model, was judged by non-inferiority: the upper exact 95 % bound of R below 1.25 (one-sided 2.5 %), with the 98.75 % bound reported beside it. At 13 materials its exact size at the margin is 0.024 (0.005 for the 98.75 % bound). Before the freeze, the power of each verdict was simulated with the analysis code itself, and the whole pipeline was run on five synthetic worlds, with an expected outcome written for each before it ran, to show which verdicts can fail (Supplementary Note 1). Three of the expectations were missed and are reported as missed, and the comparison of a free with a van't Hoff temperature dependence was made descriptive because it could not fail.

Data Availability

The pre-registration, the freeze manifest, the corrections ledger, the digitisation provenance and all results are public at `https://github.com/abdulhamidbatayhi123/ai-isotherm-thermo`; an archived release with a DOI will be cited. Digitised point lists from sources whose licence does not permit redistribution are kept locally; their provenance, fitted parameters and predictions are published.

Code Availability

All code is in the same repository and archive.

Use of AI tools

Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

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